Neurocysticercosis and oncogenesis.

Recent studies suggest that neurocysticercosis may be a risk factor for human cancer. Pathogenetic mechanisms explaining possible oncogenic effects of cysticerci include the following: (a) parasite-induced modulation of the host immune response that may be associated with loss of regulatory mechanisms implicated in the immunological surveillance against cancer; (b) transfer of genetic material from the parasite to the host, causing DNA damage and malignant transformation of host cells, and (c) chronic inflammation with liberation of nitric oxide and inhibition of tumor suppressor genes. Further research is needed to confirm the potential role of cysticercosis in the development of cancer. These studies should determine the presence of cysticercotic factors responsible for the transfer of genetic material and potential mutations in the tumor suppressor genes in proliferating astrocytes surrounding cysticercotic lesions. Additionally, the complex interaction between the immune state of the host with variable cytokine release and the presence of inflammatory cells releasing nitric oxide that cause DNA damage and impair tumor suppressive mechanisms needs to be investigated.